

Storage strategy, databases, recovery

Day 6 - Morning

Running scenario: Thara Hospitals patient portal pilot

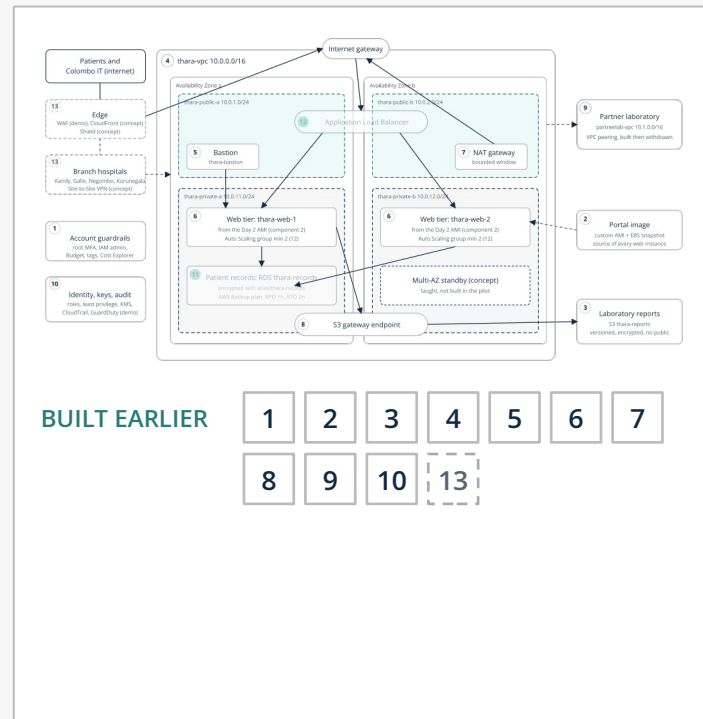
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Today

What does it cost to keep, how long to get back, and has anyone tried?

- S3 as a strategy, not a bucket - 35 min
- Managed databases - 35 min
- Recovery as a number - 30 min



CONCEPT 1

S3 as a strategy, not a bucket

Old reports in a cheaper class. An overwritten report brought back.

35 min

Storage classes: cheap to keep, or quick to fetch

5

classes to know
three of them are archives

90

days, then a cheaper class
Thara's lifecycle rule

30

days you pay for at least
in Infrequent Access

Class	Getting it back	You pay for at least
S3 Standard	At once	Nothing
S3 Standard-Infrequent Access	At once, for a charge	30 days
S3 Glacier Instant Retrieval	At once, for a higher charge	90 days
S3 Glacier Flexible Retrieval	Minutes to hours	90 days
S3 Glacier Deep Archive	Hours	180 days

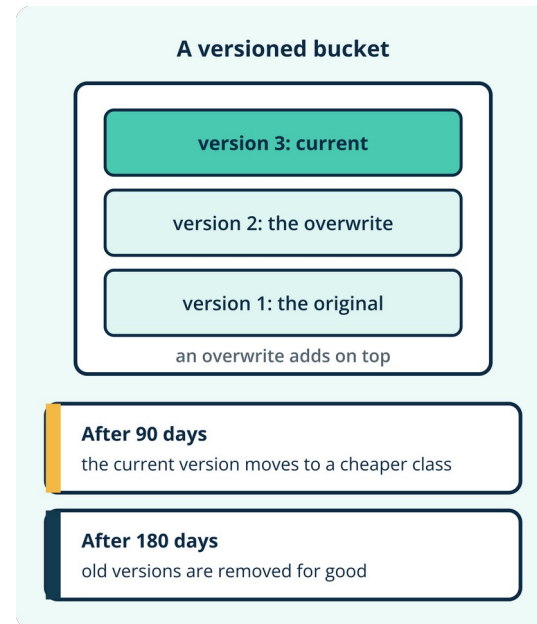
Versioning: nothing is overwritten

- **Versioning** keeps every version of an object
- An overwrite adds a new version on top
- A delete adds a marker. The versions beneath remain

COMMON MISTAKE

Versioning is a backup.

It undoes an overwrite or a delete. It does not survive the loss of the bucket.



Every version lives in the same bucket.

Lifecycle rules: storage that tidies itself

- A **lifecycle rule** moves objects to a cheaper class as they age
- It also expires old versions, or a versioned bucket grows for ever
- It acts over hours to days: configure it and verify it

T

THARA HOSPITALS

Report PDFs to Infrequent Access after 90 days; versioning so a doctor's overwritten report is recoverable.

Copies, and who can read

- **Replication** copies new objects to a second bucket. A concept today
- **Block Public Access** overrides anything that would make a bucket public
- A **bucket policy** is one document on the bucket, naming principals
- ACLs grant access object by object. Leave them disabled

YOU ALREADY KNOW THIS

Tiered storage on a SAN with archive to tape.

Object store or file system?

S3

Holds

Whole objects, each under a key.

Changed by

Put, get and delete.

Suits

Written once, read whole: a report PDF.

EFS

Holds

Files in folders, mounted by many instances.

Changed by

Editing in place.

Suits

Software that expects a shared folder.

EFS is a concept today. Thara's reports are objects.

CONCEPT 2

Managed databases

Private placement, the key from Day 5, and a tested restore.

35 min

Who does the work?

MySQL on your own EC2

You do

Operating system patches. Engine install and patches.
Backups. Failover.

You also do

Who may connect. Encryption. The data.

RDS

AWS does

Operating system patches. Engine install and patches.
Backups. Failover.

You still do

Who may connect. Encryption. The data.

The Shared Responsibility Model from Day 1, applied to one service.

Three groups, and one rule

- A **DB subnet group** lists the subnets the database may use, in two zones
- A **security group** allows the database port from the web tier's group only
- A **parameter group** holds the engine's settings
- The database is **never public**



THARA HOSPITALS

Patient records need private placement, encryption with `alias/thara-records`, and a tested restore.

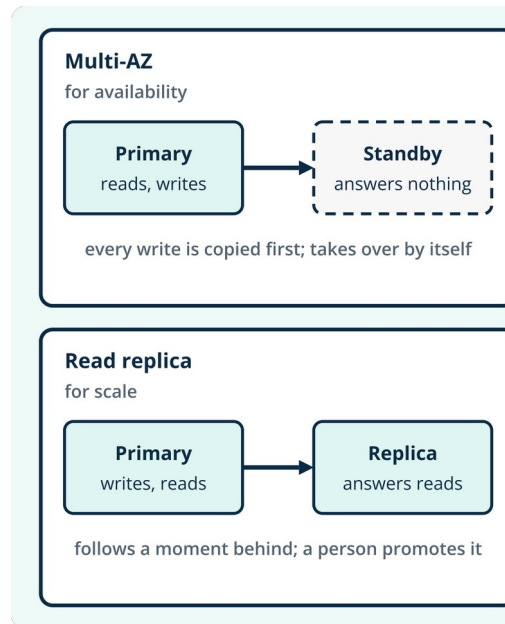
Multi-AZ for availability, replicas for scale

- **Multi-AZ**: a standby in a second zone. It takes over by itself
- The standby answers no queries while it waits
- A **read replica** answers reads at its own endpoint
- Neither is a backup. Both copy a mistake faithfully

COMMON MISTAKE

Multi-AZ improves read performance.

It does not. Replicas do.



The same two boxes with different arrows.

Backups, and one more database

- **Automated backups**: daily, kept for the days you set, gone with the instance
- A **manual snapshot**: taken when you ask, kept until you delete it
- A restore always makes a **new** database
- **DynamoDB**: key-value, no instance to size. On the map, not in the lab

YOU ALREADY KNOW THIS

A managed PBX versus running your own.

CONCEPT 3

Recovery as a number

RPO one hour, RTO two hours. Today's lab measures the RTO.

30 min

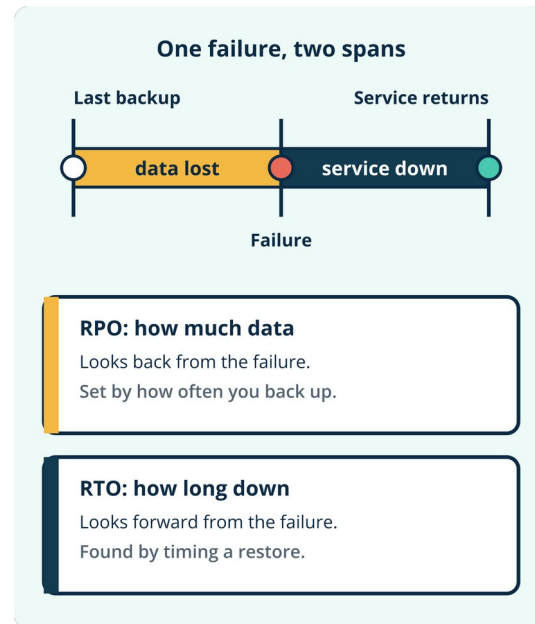
Two numbers, one failure

- **RPO**: the most data you accept losing, as time
- **RTO**: the longest the service may stay down
- Data loss is set before the failure, by how often you back up
- Downtime is found after it, by timing a restore

COMMON MISTAKE

RPO and RTO are the same number.

One is data loss. The other is downtime.



Two spans that share one moment.

AWS Backup: one plane across services

- A **plan**: what is backed up, how often, and for how long
- A **vault**: where the backups are kept
- **Retention**: when a backup is removed for you
- An **on-demand backup**: one taken now, under the same rules

Backup and restore is the first recovery pattern: the cheapest, and the slowest.

YOU ALREADY KNOW THIS

The tape in the third-floor cupboard, and the day nobody could read it.

A restore never tested is not a backup

Restored, timed, checked

The database is deleted on purpose and brought back from its snapshot. The rows are read, and the minutes are written down.

A measured RTO to set against two hours.

Backed up, never restored

Backups run every night for years. On the day they are needed the copy is incomplete, or lands in the wrong network, or takes six hours.

Three days on paper.

This afternoon you do the first of these.



IT

HEAD OF IT

One hour of data, two hours of downtime. Those are the numbers I gave the board. Until someone restores the records with a clock running, the second one is a guess.

The Head of IT, on the patient records. Today's lab measures the RTO.

Demonstration: Failover, lifecycle, plan



1. A Multi-AZ database in the instructor's account. What a failover would do, described from the console
2. A lifecycle rule, and an object marked to move. Why it cannot be watched today
3. A Backup plan and a vault: the schedule and retention fields

Nothing here is triggered live. This afternoon the restore is.

Activity: classify Thara's data, 35 minutes



1. In pairs, complete the classification table for Thara's six kinds of data
2. For each: the storage service, the protection, and the recovery target
3. Compare with the reference table
4. Argue the two rows on which most pairs disagree

The classification table

3

classes

ephemeral, recoverable, critical

6

kinds of data

one row each

2

rows to argue

where pairs disagree

Data	Class	Protection	Recovery target
Web server and application	...	...	...
Laboratory report PDFs	...	...	...
Patient records database	...	...	...
Audit logs	...	...	...
Flow logs	...	...	...
Portal session data	...	...	...

Common mistakes



COMMON MISTAKE

Versioning is a backup.

It protects against overwrite and delete, not against the loss of the bucket.

COMMON MISTAKE

Multi-AZ improves read performance.

It does not. Replicas do.

COMMON MISTAKE

RPO and RTO are the same number.

One is data loss. The other is downtime.

Before the next session



READ

Wittig and Wittig, AWS in Action

The chapter on RDS.

SELF-CHECK

CLF-C02 bank

Domain 4 set B, 10 items.

PRACTICE

PQ6

A backup and recovery design against a stated RPO and RTO, with cost.

Evidence Pack 6 follows this afternoon's lab.

Questions & Discussion



Next: this afternoon the records are built, deleted and brought back.



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